

PROPOSITIONAL LOGIC

Discrete Mathematics

I.2 PROPOSITIONAL EQUIVALENCES

- Tautology
 - A compound proposition that is always true no matter what the truth values of the propositions that occur in it is called tautology.
- Contradiction
 - A compound proposition that is always false is called contradiction.
- Logical Equivalence
 - Compound propositions that have the same truth values in all possible cases are called logically equivalent.

LOGICAL EQUIVALENCE

1. Prove that $\neg (p \vee q)$ and $\neg p \wedge \neg q$ are logically equivalent.

2. Prove or disprove the following equivalence using truth table:

$$p \rightarrow q \equiv \neg p \vee q$$

$$p \rightarrow q \equiv \neg p \vee q???$$

p	q	$p \rightarrow q$	$\neg p$	$\neg p \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

EXERCISE

- Prove that $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent.

ANOTHER WAY TO VERIFY EQUIVALENCE

- We remember some important equivalences
- We use them to verify any given equivalence

IMPORTANT LOGICAL EQUIVALENCES

IDENTITY

$$p \wedge T \equiv p$$

$$p \vee F \equiv p$$

DOMINATION

$$p \vee T \equiv T$$

$$p \wedge F \equiv F$$

IMPORTANT LOGICAL EQUIVALENCES

IDEMPOTENT

$$p \wedge p \equiv p$$

$$p \vee p \equiv p$$

DOUBLE NEGATION

$$\neg(\neg p) \equiv p$$

IMPORTANT LOGICAL EQUIVALENCES

COMMUTATIVE

$$p \vee q \equiv q \vee p$$

$$p \wedge q \equiv q \wedge p$$

ASSOCIATIVE

$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$

$$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

IMPORTANT LOGICAL EQUIVALENCES

DISTRIBUTIVE

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

IMPORTANT LOGICAL EQUIVALENCES

DE MORGAN'S LAW

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

IMPORTANT LOGICAL EQUIVALENCES

ABSORPTION

$$p \vee (p \wedge q) \equiv p$$

$$p \wedge (p \vee q) \equiv p$$

NEGATION

$$p \vee \neg p \equiv T$$

$$p \wedge \neg p \equiv F$$

IMPORTANT LOGICAL EQUIVALENCES

CONDITIONAL

$$p \rightarrow q \equiv \neg p \vee q$$

CONTRAPOSITIVE

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

BICONDITIONAL

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

EXAMPLE: PROVE THAT

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

Rosen, page 26, example 6

EXERCISE: PROVE
THAT $\neg(p \rightarrow q) \equiv p \wedge \neg q$

$$\neg(p \rightarrow q)$$

$$\equiv \neg(\neg p \vee q) \quad [\text{Conditional}]$$

$$\equiv \neg(\neg p) \wedge \neg q \quad [\text{De Morgan's Law}]$$

$$\equiv p \wedge \neg q \quad [\text{Double Negation}]$$

EXAMPLE: PROVE THAT

$$\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$$

Rosen, page 26, example 7

EXERCISE: PROVE THAT $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$

$$\neg(p \vee (\neg p \wedge q))$$

$$\equiv \neg(p) \wedge \neg(\neg p \wedge q)$$

$$\equiv \neg p \wedge [\neg(\neg p) \vee \neg q]$$

$$\equiv \neg p \wedge (p \vee \neg q)$$

$$\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q)$$

$$\equiv F \vee (\neg p \wedge \neg q)$$

$$\equiv (\neg p \wedge \neg q)$$

EXAMPLE: PROVE THAT $(p \wedge q) \rightarrow (p \vee q)$ IS A TAUTOLOGY

Rosen, page 27, example 8

EXERCISE: PROVE
THAT $(p \wedge q) \rightarrow (p \vee q)$ IS A
TAUTOLOGY

$$\begin{aligned}(p \wedge q) &\rightarrow (p \vee q) \\ &\equiv \neg(p \wedge q) \vee (p \vee q) \\ &\equiv (\neg p \vee \neg q) \vee (p \vee q) \\ &\equiv (\neg p \vee p) \vee (q \vee \neg q) \\ &\equiv T \vee T \\ &\equiv T\end{aligned}$$

EXAMPLE: PROVE THAT

$$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

Rosen, page 29, exercise 22

$$\begin{aligned}(p \rightarrow q) \wedge (p \rightarrow r) \\ &\equiv (\neg p \vee q) \wedge (\neg p \vee r) \\ &\equiv \neg p \vee (q \wedge r) \\ &\equiv p \rightarrow (q \wedge r)\end{aligned}$$

EXERCISE: $(p \rightarrow q) \wedge$
 $(p \rightarrow r) \equiv p \rightarrow (q \wedge r)$

EXERCISE: PROVE THAT $\neg p \rightarrow$
 $(q \rightarrow r) \equiv q \rightarrow (p \vee r)$

Rosen, page 29, exercise 26

EXERCISE: PROVE
THAT $\neg p \rightarrow (q \rightarrow r) \equiv$
 $q \rightarrow (p \vee r)$

Rosen, page 29, exercise 26

$$\begin{aligned}\neg p \rightarrow (q \rightarrow r) \\ &\equiv \neg(\neg p) \vee (q \rightarrow r) \\ &\equiv p \vee (q \rightarrow r) \\ &\equiv p \vee (\neg q \vee r) \\ &\equiv p \vee \neg q \vee r \\ &\equiv \neg q \vee p \vee r \\ &\equiv \neg q \vee (p \vee r) \\ &\equiv q \rightarrow (p \vee r)\end{aligned}$$

EXERCISE: PROVE THAT

$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

Rosen, page 28, exercise 16

$$\neg(p \leftrightarrow q)$$

$$\equiv \neg((p \rightarrow q) \wedge (q \rightarrow p))$$

$$\equiv \neg(p \rightarrow q) \vee \neg(q \rightarrow p)$$

$$\equiv \neg(\neg p \vee q) \vee \neg(\neg q \vee p)$$

$$\equiv (\neg(\neg p) \wedge \neg q) \vee (\neg(\neg q) \wedge \neg p)$$

$$\equiv (p \wedge \neg q) \vee (q \wedge \neg p)$$

$$\equiv (p \vee q) \wedge (p \vee \neg p) \wedge (\neg q \vee q) \wedge (\neg q \vee \neg p)$$

$$\equiv (p \vee q) \wedge (\neg p \vee \neg q)$$

$$\equiv (q \vee p) \wedge (\neg p \vee \neg q)$$

$$\equiv (\neg(\neg q) \vee p) \wedge (\neg p \vee \neg q)$$

$$\equiv (\neg q \rightarrow p) \wedge (p \rightarrow \neg q)$$

$$\equiv (p \rightarrow \neg q) \wedge (\neg q \rightarrow p)$$

$$\equiv (p \leftrightarrow \neg q)$$

[Biconditional]

[De Morgan's]

[Conditional]

[De Morgan's]

[Double Negation]

[Distributive]

[Negation, Identity]

[Commutative]

[Double Neg]

[Implication]

[Commutative]

[Biconditional]

EXAMPLE: PROVE THAT $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$ IS A TAUTOLOGY

Rosen, page 29, exercise 29

EXAMPLE: PROVE THAT
 $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$
IS A TAUTOLOGY

Rosen, page 29, exercise 29

$$\begin{aligned} & (p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r) \\ \equiv & \neg((p \rightarrow q) \wedge (q \rightarrow r)) \vee (p \rightarrow r) \\ \equiv & \neg(p \rightarrow q) \vee \neg(q \rightarrow r) \vee (p \rightarrow r) \\ \equiv & \neg(\neg p \vee q) \vee \neg(\neg q \vee r) \vee \neg p \vee r \\ \equiv & (p \wedge \neg q) \vee (q \wedge \neg r) \vee \neg p \vee r \\ \equiv & (p \wedge \neg q) \vee \neg p \vee (q \wedge \neg r) \vee r \\ \equiv & ((p \vee \neg p) \wedge (\neg q \vee \neg p)) \vee \\ & ((q \vee r) \wedge (\neg r \vee r)) \\ \equiv & (T \wedge (\neg q \vee \neg p)) \vee ((q \vee r) \wedge T) \\ \equiv & (\neg q \vee \neg p) \vee (q \vee r) \\ \equiv & \neg p \vee \neg q \vee q \vee r \\ \equiv & \neg p \vee T \vee r \\ \equiv & T \end{aligned}$$